

Lecture 1

Why deep learning now – and the foundations beneath it

STAT 940 · Deep Learning

Learning goals and reading map

By the end of this lecture, you should be able to:

- Explain why deep learning became a general-purpose technology
- Distinguish major machine-learning task types
- Describe model, hypothesis class, and score/loss
- Trace the path from a biological neuron to a feedforward network

Elements of Deep Learning

Core reading

Chapter 1, Sections 1–5
Chapter 2, Sections 2.1–2.9

Recent 2024–2026 examples are supplementary and postdate the book.

Why deep learning now?

Deep learning crossed five frontiers

1

GENERATE

text, image, audio

2

SIMULATE

interactive worlds

3

REASON

math, planning

4

ACT

robots, agents

5

DISCOVER

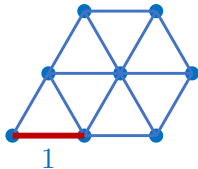
biology, weather,
materials

The striking change is **breadth**: one family of learning methods now reaches across all five.

Mathematics frontier: unit distances

How many pairs of points can be exactly one unit apart?

- A long-standing Erdős conjecture predicted a specific upper bound for points in the plane.
- The challenge is not calculation alone; it is discovering a construction that no one had seen.



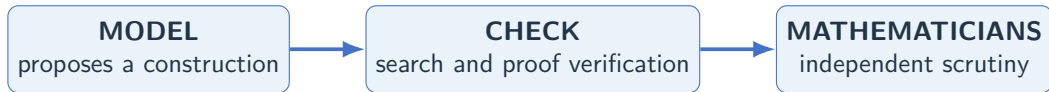
A geometric construction can overturn a universal claim.

May 2026: AI found a counterexample

DISPROVED

A long-standing Erdős conjecture in discrete geometry

The model produced an infinite family with a polynomial improvement – not merely one numerical exception.



The lesson: discovery can emerge from a loop of generation, computation, and verification.

IMO 2025: gold-medal-standard reasoning

35 / 42

officially graded points

Gold

medal threshold

5 / 6

problems solved

4.5 h

competition time

Solutions were written in natural language and evaluated under the same time limit as the human competition.

World models: prompt to navigable world

Genie 3

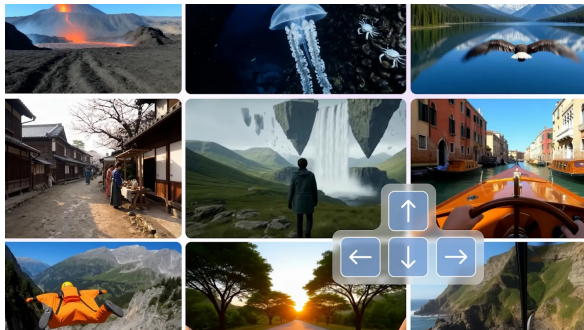
- Generates dynamic worlds from text
- Supports real-time navigation and interaction
- Maintains consistency for a few minutes

720p

resolution

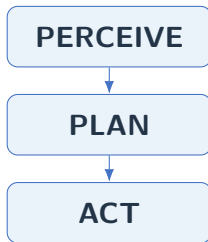
24 fps

real-time generation



Embodied agents: perceive, plan, and act

Gemini Robotics 1.5



The model links vision, language, reasoning, and motor control for complex multi-step physical tasks.

One interaction can now cross modalities

PROMPT + CONTEXT



The interface is becoming less about a single output type and more about a **shared learned representation**.

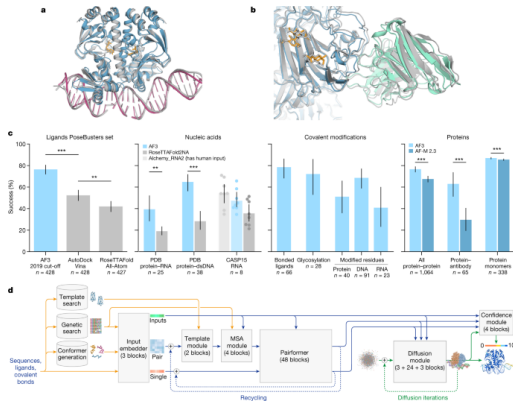
Scientific discovery: molecular interactions

AlphaFold 3

A unified model for complexes containing:

- proteins
- nucleic acids
- small molecules and ions
- modified residues

The target is no longer one isolated structure; it is the molecular interaction system.



Scientific discovery: global weather

GenCast

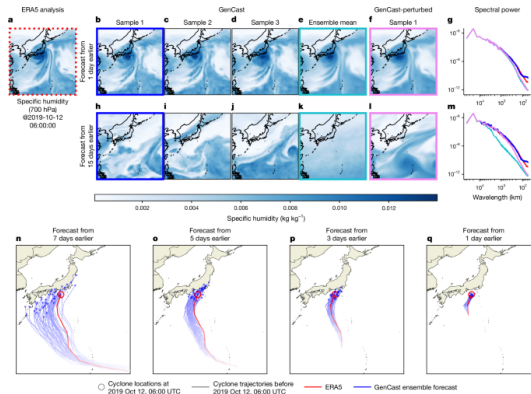
15
days

global
ensemble
forecast

8
min

generation
time

Better than the operational ENS system
on **97.2% of 1,320 evaluated targets.**



Scientific discovery: new materials

GNoME

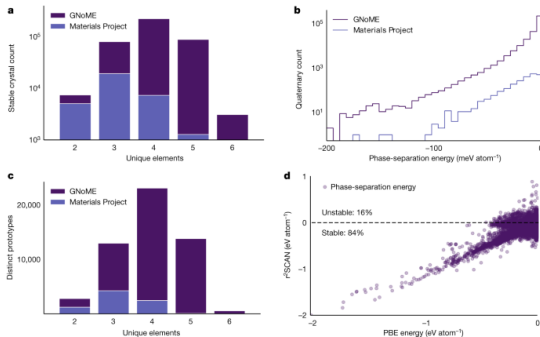
2.2 M

stable structures

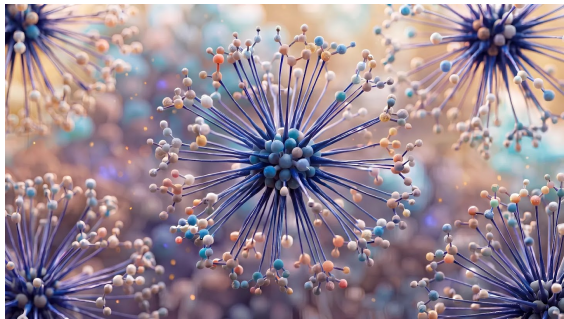
381 k

new structures on the
updated convex hull

Deep learning turns an enormous combinatorial search space into a ranked set of candidates for computation and experiment.



Scientific discovery: hypothesis generation



AI co-scientist

A multi-agent system that repeatedly:

- ① generates hypotheses
- ② debates and ranks them
- ③ uses tools and evidence
- ④ refines the research plan

The output is not just an answer – it is a structured scientific search process.

What remains genuinely hard?

RELIABILITY

Know when the
model is wrong

CAUSALITY

Reason beyond
correlations

CONTINUAL LEARNING

Learn without
forgetting

EMBODIMENT

Act safely
in the
physical world

EVALUATION

Measure
capability
and risk

Impressive demonstrations are not the same as a complete theory – or a dependable system.

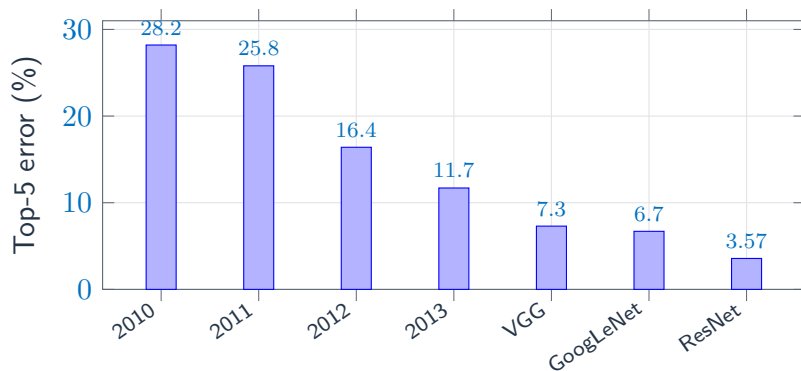
How did we get here?

A compact history of deep learning



The ideas are old; the inflection came from **data, compute, optimization, and scalable architectures.**

ImageNet changed computer vision



2012

AlexNet sharply reduced ImageNet error and accelerated the shift toward deep convolutional networks.

What is machine learning?

Classical statistics

Infer structure and uncertainty from data, often under explicit modeling assumptions.



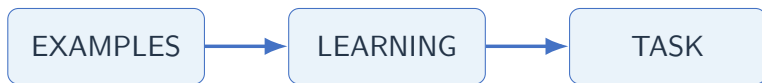
Machine learning

Learn predictive or descriptive functions from examples, often at large scale.

The boundary is not absolute; the emphasis shifts toward algorithms that improve from data.

Learning from examples

Machine learning teaches a computer through examples rather than an exhaustive list of explicit rules.



Examples are especially useful when the desired behavior would be difficult to program directly.

Three broad learning settings

Supervised learning

Learn from input–output examples.

- Classification
- Regression

Unsupervised learning

Find structure without externally supplied targets.

- Clustering
- Dimensionality reduction

Reinforcement learning

Learn actions from rewards through interaction with an environment.

Self-supervised learning uses supervised objectives with targets constructed from the data itself.

Classification

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n, \quad \mathbf{x}_i \in \mathbb{R}^d, \quad y_i \in \mathcal{Y}, \quad |\mathcal{Y}| < \infty.$$

Find a function

$$f : \mathbb{R}^d \rightarrow \mathcal{Y}$$

so that for a new input $\mathbf{x}$, the prediction is $f(\mathbf{x})$.

Regression

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n, \quad \mathbf{x}_i \in \mathbb{R}^d, \quad y_i \in \mathbb{R}.$$

Find a function

$$f : \mathbb{R}^d \rightarrow \mathbb{R}$$

so that for a new input $\mathbf{x}$, the predicted response is $f(\mathbf{x})$.

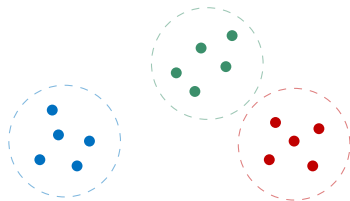
Clustering

$$\mathcal{D} = \{\mathbf{x}_i\}_{i=1}^n, \quad \mathbf{x}_i \in \mathbb{R}^d.$$

Find an assignment

$$f : \mathbb{R}^d \rightarrow \{1, \dots, K\}$$

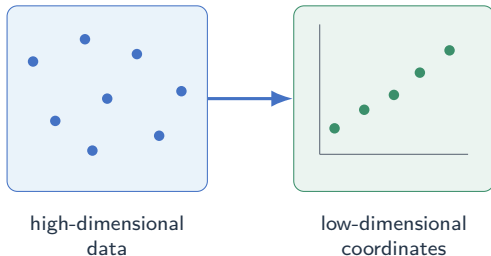
such that similar observations receive the same cluster label.



Dimensionality reduction

Represent high-dimensional data with fewer coordinates while preserving useful structure.

- Examples: PCA, manifold learning, autoencoders
- Uses: visualization, compression, denoising, features



Beyond the classical four

Adapting with limited examples

- Zero-shot learning
- One-shot and few-shot learning
- In-context learning

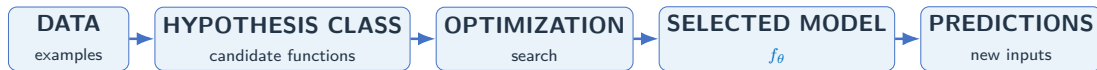
Learning across tasks and time

- Transfer learning
- Multi-task learning
- Meta-learning
- Continual learning

These regimes do not replace the classical task types.

They describe how a model transfers, adapts, or continues learning across tasks.

The general learning procedure



A score or loss tells optimization which candidate is better.

Model = hypothesis class

- Linear models and generalized linear models
- Trees, kernels, and support-vector machines
- Probabilistic graphical models
- Deep networks with learned representations

A model class specifies the candidate functions available to learning.

Score / loss criterion

$$\theta^* = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n \mathcal{L}(f_{\theta}(\mathbf{x}_i), y_i).$$

LOSS

What counts as error?

OPTIMIZATION

How do we search?

GENERALIZATION

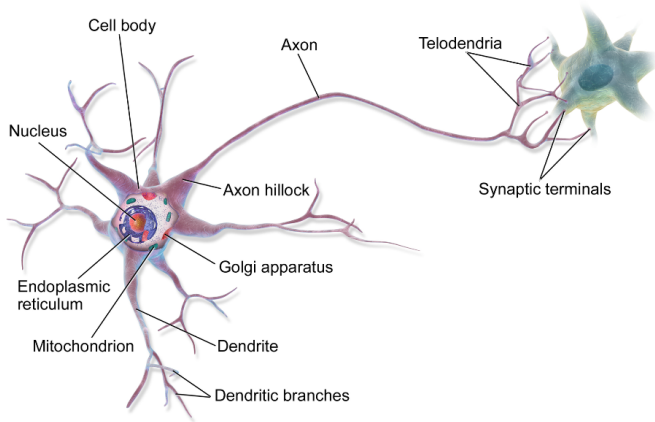
Will it work on new data?

Deep structures are hypothesis classes



Architecture is an **inductive bias**: it defines which functions are easy for learning to express.

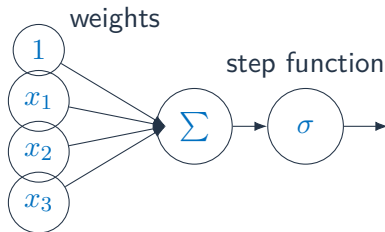
Biological inspiration



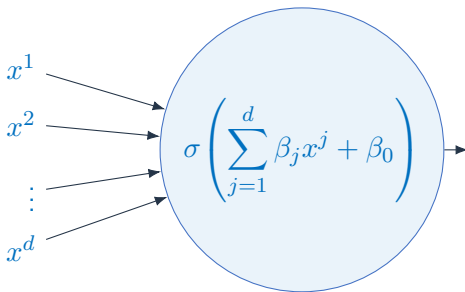
1958: the perceptron

1958

- Frank Rosenblatt developed the perceptron in 1958.
- It is one of the earliest artificial-neuron models, inspired by biological neurons.



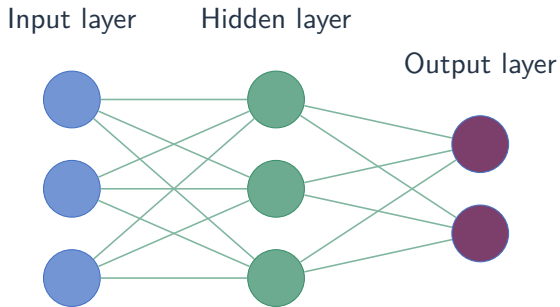
The artificial neuron



Simple perceptron

x^j is the j th feature, β_j is its weight, and β_0 is the bias. The bias shifts the decision boundary $\beta^\top \mathbf{x} + \beta_0 = 0$.

From neurons to layers

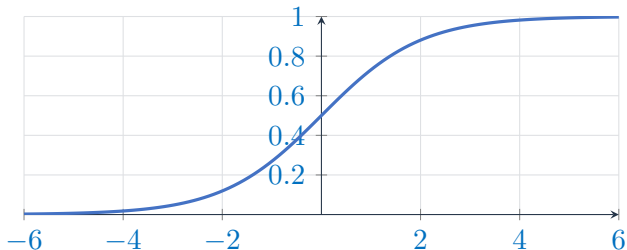


Neurons are connected by weighted edges; a layer transforms one representation into the next.

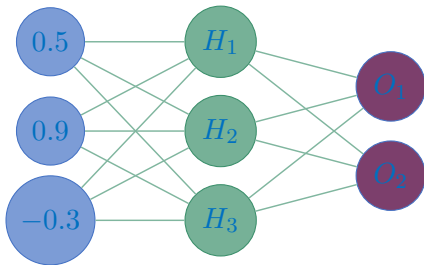
Activation functions

Activation functions are applied to the weighted input at each neuron. A classical example is the sigmoid:

$$\sigma(t) = \frac{1}{1 + e^{-t}}.$$



Inference: one layer at a time



$$\mathbf{w}_{H_1} = (1, -2, 2)$$

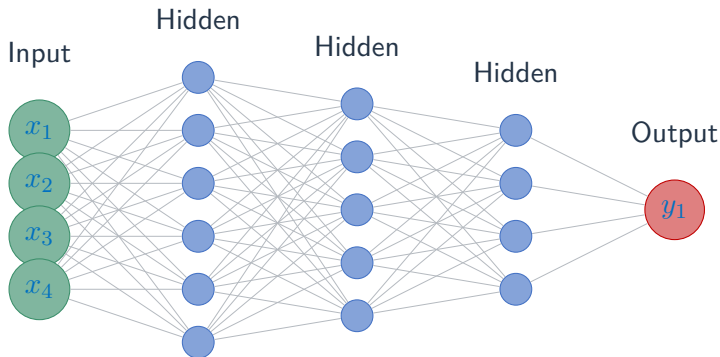
$$\mathbf{w}_{H_2} = (2, 1, -4)$$

$$\mathbf{w}_{H_3} = (1, -1, 0)$$

$$\mathbf{w}_{O_1} = (-3, 1, -3)$$

$$\mathbf{w}_{O_2} = (0, 1, 2)$$

Feedforward computation



A feedforward network composes layerwise transformations from input to output.

Takeaways and reading

Four ideas to keep

- ① Deep learning is now a general-purpose method
- ② Learning starts with tasks, examples, and a criterion
- ③ Architecture defines a hypothesis class
- ④ A feedforward network composes simple learned transformations

Read before Lecture 2

Essential

Chapter 1, Sections 2, 3, 5
Chapter 2, Sections 2.3–2.9

For context

Chapter 1, Sections 1, 4
Chapter 2, Sections 2.1–2.2

The slides are the teaching baseline; the book is the supporting reference.